# Spectral-Omics: Phasor Vertices, Omics Connectome Harmonics, and the Compartment Coupling Matrix

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# Abstract

Gene expression, chromatin state, and metabolite abundance are noisy, high dynamic-range, and intrinsically oscillatory — properties that the linear embeddings routinely used to summarise omics data (principal-component analysis, non-negative matrix factorisation) discard because a real-valued score cannot carry a phase. We introduce *Spectral-Omics*, a phase-aware graph-spectral framework that encodes each molecular feature as a complex *phasor vertex* $\psi_i = r_i e^{i\theta_i}$, assembles a Hermitian *Omics Connectome Matrix* (OCM) $H_{ij} = c_{ij}\,\psi_i\psi_j^*$, and diagonalises it into *omics harmonics* (collective normal modes). The construction is Hermitian, gauge-invariant under a global phase shift, and passes a suite of algebraic consistency checks at machine precision. On a lung-adenocarcinoma cohort (GSE10072) the phasor spectral indicators separate tumour from normal tissue at the ceiling of the metric (held-out AUROC 0.998), but a leakage-controlled ablation shows the phase factor adds no discriminative power over an amplitude-only baseline (AUROC 0.997; paired $p = 0.80$) or over PCA and WGCNA eigengenes, a result that also holds off-ceiling on a small-sample learning curve. The phase is informative where timing is the signal: on a mouse-liver circadian time-course (GSE11923, 48 hourly samples) the phasor recovers core-clock acrophases (circular correlation 0.997 against cosinor; correct Bmal1–Per antiphase), and a day-1-select / day-2-test protocol shows the collective spectral rhythm persists out-of-sample (participation-ratio $R^2 = 0.68$, orthogonal to and exceeding the mean-expression rhythm). We show that the default construction's gradient edge-phase gives zero cycle flux and is isospectral with the real coupling, and introduce a non-zero-flux magnetic OCM variant that makes the phase encode direction of flow when a directed or signed coupling is available. Enrichment analysis shows that the leading harmonic is an expression-magnitude summary and that only sub-leading modes carry annotation-supported biology; we therefore treat the compartment readout as a descriptive decomposition rather than a validated classifier. Spectral-Omics is provided as an installable, tested package and is designed as an interpretable spectral lens on molecular coordination and timing.



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## I. INTRODUCTION

Modern systems biology measures the cell in ever finer molecular detail — transcriptome, accessible genome, proteome, metabolome — yet the dominant analysis paradigm still reduces each such matrix to a low-dimensional Euclidean embedding and interprets the axes of that embedding. Principal-component analysis, its sparse and non-negative variants, and the eigengene/eigenarray decomposition of Alter *et al.* [1] are all of this form, and they have been extraordinarily productive: hierarchical clustering of expression [2] and weighted co-expression networks [3] organise genome-scale data into interpretable modules. These methods share one structural assumption, however — that a feature's state is a real-valued magnitude — and that assumption discards a property which is central to much of cell biology: *oscillation*.

### A. Omics data are oscillatory, and magnitude-only methods hide it

A large fraction of the transcriptome and proteome is rhythmic. The circadian clock alone drives on the order of 40% of the protein-coding genome in a tissue-specific manner [4, 5]; the cell cycle imposes a second periodicity [6]; and metabolic and redox rhythms persist even when transcription is arrested [7]. Detecting which genes oscillate is by now standard — Fourier and nonparametric periodicity tests such as JTK_CYCLE [8], Lomb–Scargle periodograms [9], and cosinor regression [10] are routine. What remains awkward is representing that rhythmicity *inside* the downstream dimensionality reduction. A real-valued embedding can encode that two gene programmes are both rhythmic, but not that they are a quarter-cycle out of phase, because relative phase is exactly the degree of freedom a magnitude throws away.

### B. A phase is a natural coordinate — and has precedent

Encoding a measurement as a complex number $re^{i\theta}$ so that its phase becomes a first-class coordinate is not new to biology. The *phasor* representation of fluorescence-lifetime imaging [11, 12] maps each pixel's decay to a point in the complex plane and has become a standard, fit-free tool; the "cell phasor" distinguishes metabolic states of stem cells directly from that geometry [13]. The common lesson is that once data carry a phase, mixtures, transitions, and relative timing become geometric statements. We adopt the same stance for molecular abundance: represent feature $i$ as a phasor vertex $\psi_i = r_i e^{i\theta_i}$ whose magnitude encodes abundance and whose phase encodes position in a shared oscillatory cycle.

### C. From phasors to a Hermitian graph and its harmonics

A population of $N$ features is then a field of phasors on a graph, and the natural object of study is not their Euclidean covariance but a Hermitian coupling operator

$$H_{ij} \;=\; c_{ij}\, \psi_i \psi_j^* \;=\; c_{ij}\, r_i r_j\, e^{i(\theta_i - \theta_j)}, \tag{1}$$

whose off-diagonal phase factor $e^{i(\theta_i-\theta_j)}$ measures the *relative* phase of two features, weighted by a co-expression strength $c_{ij}$. Hermitian and complex-valued graph representations are well studied in spectral graph theory — the Hermitian adjacency matrix of a directed graph [14] and the broader programme of graph signal processing [15, 16] show that a Hermitian operator on a graph has a real spectrum and an orthogonal eigenbasis that generalises the Fourier modes of a line to an arbitrary network. Eigenmodes of biological graphs are already used as functional coordinates: connectome harmonics organise large-scale brain activity [17], spectral clustering and modularity partition interaction networks [18, 19], and diffusion-map eigenvectors chart single-cell differentiation [20]. Diagonalising eq. (1) places molecular data in the same setting: its eigenvectors, which we call *omics harmonics*, are collective normal modes of the molecular network, and its eigenvalues rank how much coordinated variation each mode carries.

### D. Contributions

This paper makes four contributions.

1. We define the amplitude-weighted Omics Connectome Matrix (section II) and prove that the phasor amplitudes are what make its spectrum sample-specific, while the gradient relative-phase construction keeps it Hermitian and invariant under a global gauge shift — and, as a direct corollary, has zero cycle flux and is isospectral with the real coupling.

2. We benchmark the framework on two public datasets (section IV). On a tumour/normal cohort the phasor indicators match strong linear baselines but the phase adds no discrim-

inative power; on a circadian time-course the phase is informative, recovering clock-gene acrophases and an out-of-sample collective rhythm distinct from mean expression.

3. We position the construction against the magnetic-Laplacian lineage for directed and signed graphs [14, 21–25] and introduce a non-zero-flux magnetic OCM variant (section II C) that makes the phase do topological work when a directed or signed coupling is available.

4. We test the interpretive layer against external annotation (section IV E); the leading mode is an expression-magnitude summary and the compartment layer is not anchored to ground-truth annotation, so we present it as a descriptive decomposition. The whole pipeline is an installable, tested package with an algebraic consistency suite that holds at machine precision.

## II. THEORETICAL FRAMEWORK

We fix notation: $N$ molecular features (genes, proteins, peaks, or metabolites), $S$ samples, and, for time-course data, $T$ time points. An omics matrix is $X \in \mathbb{R}^{S\times N}$, non-negative for count-like assays. Bold lowercase denotes vectors, bold uppercase matrices; $\dagger$ is the conjugate transpose and $\langle\cdot\rangle$ an average.

### A. Phasor-vertex encoding

Each feature $i$ in a given sample is encoded as a complex *phasor vertex* $\psi_i = r_i e^{i\theta_i} \in \mathbb{C}$, with a non-negative amplitude $r_i \in [0, 1]$ and a phase $\theta_i \in (-\pi, \pi]$. The phase uses a bounded, outlier-robust tanh encoding: for a feature column $x$ across samples,

$$\theta \;=\; \pi \tanh\!\left(\frac{\log(1+x)-\mu}{\sigma+\varepsilon}\right), \tag{2}$$

where $\mu$ and $\sigma$ are the mean and standard deviation of $\log(1+x)$ across samples and $\varepsilon = 10^{-8}$. The $\log(1+\cdot)$ compresses the dynamic range, per-feature standardisation removes scale, and $\pi\tanh(\cdot)$ maps the standardised value onto the phase circle while saturating — rather than wrapping — outliers, so a single aberrant sample cannot fold a feature onto the opposite phase. For assays already on a logarithmic footing (chromatin accessibility, methylation $\beta$-values) the log step is omitted. The amplitude is a per-feature min–max of $\log(1 + x)$

into $[0,1]$ (mode `expression`) or the amplitude of the leading periodic component in a time-course (mode `rhythm`). Global phase alignment across the population is summarised by the Kuramoto order parameter [26]

$$R \;=\; \left|\tfrac{1}{N}\textstyle\sum_n e^{i\theta_n}\right| \in [0,1], \tag{3}$$

with $R \to 1$ for a phase-aligned population and $R \to 0$ for a dispersed one.

### B. The Omics Connectome Matrix

Given a phasor vector $\psi \in \mathbb{C}^N$ for one slice and a non-negative symmetric coupling matrix $c_{ij}$, the OCM is

$$H_{ij} \;=\; c_{ij}\,\psi_i\psi_j^*, \qquad H \;=\; \mathrm{diag}(\psi)\,C\;\mathrm{diag}(\psi)^\dagger. \tag{4}$$

The coupling $c_{ij}$ is a feature–feature affinity estimated once across all samples: by default the absolute Pearson correlation of feature profiles, with a soft-thresholded (scale-free, WGCNA-style [3]) variant $|\mathrm{corr}|^\beta$ and a prior-adjacency variant (from a curated regulatory or interaction network) also available. $H$ is Hermitian by construction, so its eigenvalues are real — the prerequisite for reading its eigenvectors as harmonics, exactly as for the Hermitian adjacency operators of spectral graph theory [14, 16].

**Proposition 1** (Amplitude weighting makes the spectrum informative)**.** *Let $C$ be fixed. If the amplitudes are discarded ($r_i \equiv 1$), then* $\mathrm{diag}(\psi) = \mathrm{diag}(u)$ *with $u_i = e^{i\theta_i}$ is* unitary*, so $H = \mathrm{diag}(u)\,C\;\mathrm{diag}(u)^\dagger$ is a unitary similarity of $C$ and* $\mathrm{spec}(H) = \mathrm{spec}(C)$ *for every slice. The spectrum then carries no sample-specific information. With the amplitudes restored,* $\mathrm{diag}(\psi)$ *is non-unitary and eq. (4) is a* congruence *of $C$; by Sylvester's law of inertia the signature is preserved but the eigenvalues move with the sample.*

Proposition 1 is the central modelling choice of Spectral-Omics: the amplitude-weighted form is the default, and the pure-phase form is retained only as a diagnostic for isolating phase geometry. A global gauge shift $\theta_i \to \theta_i + \alpha$ (equivalently $\psi \to \psi e^{i\alpha}$) leaves the outer product $\psi_i\psi_j^*$ — and therefore $H$ and its whole spectrum — invariant, so only physically meaningful relative phases enter, while the amplitudes $r_i$ are untouched by the gauge. Because $C$ is a similarity/affinity matrix with non-negative entries, $\mathrm{tr}\,H = \sum_i c_{ii} r_i^2$ is real and the spectrum is bounded through the Perron–Frobenius eigenvalue of $|H|$.

### C. Zero cycle flux, and how to make the phase do topological work

The default OCM of eq. (4) assigns each edge the phase $\arg H_{ij} = \theta_i - \theta_j$. This is a *pure gradient* of the vertex potential $\theta$: around any cycle the edge phases telescope,

$$(\theta_i - \theta_j) + (\theta_j - \theta_k) + (\theta_k - \theta_i) \equiv 0 \pmod{2\pi}, \tag{5}$$

so the holonomy (cycle flux) vanishes identically. Equivalently, writing $u_i = e^{i\theta_i}$, the pure-phase form $H = \mathrm{diag}(u)\, C\, \mathrm{diag}(u)^\dagger$ is a unitary *congruence* of the real coupling $C$, and by Proposition 1 it is isospectral with $C$ for every slice. We stress that this is a deliberate and interpretable design choice, not an oversight: the gradient phase guarantees gauge invariance (section II B), keeps the spectrum a transparent function of amplitude $\times$ relative phase, and makes the eigenvectors readable as standing harmonics of a fixed affinity graph. The price is that in the gradient construction the *phase carries no topological information* — it is gauge-equivalent to a phaseless real operator, and cannot, by itself, encode a direction of flow around a loop.

*a. The magnetic-Laplacian lineage.* The natural way to make a phase do topological work is to let the edge phase carry an *antisymmetric* part that is not a coboundary. This is precisely the device behind the Hermitian / magnetic-Laplacian family for directed and mixed graphs. Guo and Mohar's Hermitian adjacency matrix assigns a digraph edge the phase $\pm i$ according to its orientation, giving a Hermitian operator with a real spectrum that distinguishes a digraph from its reverse [14]. The magnetic Laplacian — a complex deformation of the combinatorial Laplacian in which a "charge" parameter $q$ rotates each edge by $e^{\pm i2\pi q}$ — was used by Fanuel and co-workers to embed and cluster directed networks via magnetic eigenmaps [24, 27], and Furutani et al. built a full graph-signal-processing framework on the same Hermitian Laplacian [23]. MagNet made the charged magnetic Laplacian the propagation operator of a spectral graph neural network, showing that the phase $q$ "attunes spectral information to variation among directed cycles" [21]; MSGNN extended this to *signed* directed graphs through a magnetic signed Laplacian [22]. Böttcher and Porter placed the whole construction in the broader setting of networks with complex (Hermitian) edge weights, where the imaginary part encodes rotation/orientation between nodes [25]. In every case the power of the method comes from a *non-zero cycle flux*: an Aharonov–Bohm-type phase around loops that a gradient can never produce.

*b. A non-zero-flux* OCM *variant.* We therefore introduce, as an additive companion to eq. (4), a magnetic OCM whose edge phase carries a genuinely antisymmetric orientation $A_{ij} = -A_{ji}$,

$$H^{\mathrm{mag}}_{ij} \;=\; r_i r_j\, c_{ij}\, \exp\!\big(i\, 2\pi q\, A_{ij}\big), \qquad \Phi_{ijk} \;=\; 2\pi q\,(A_{ij} + A_{jk} + A_{ki}), \tag{6}$$

where $c_{ij} \geq 0$ is the same symmetric coupling as before, $r_i = |\psi_i|$ are the phasor amplitudes, and $q$ is a charge parameter. Because $A$ is antisymmetric but not a coboundary, the triangle flux $\Phi_{ijk}$ is generically non-zero and the operator is no longer a congruence of $C$: the spectrum now *depends on the flux.* $H^{\mathrm{mag}}$ remains Hermitian by construction (we verify $\|H^{\mathrm{mag}} - H^{\mathrm{mag}\dagger}\| = 0$ to machine precision), so the harmonic interpretation of section II B is preserved, and vertex-gauge shifts $\psi \to \psi e^{i\alpha}$ still leave the spectrum invariant — only the flux, a gauge-invariant loop quantity, moves it. The antisymmetric part $A$ is estimated from a directed or signed relationship on a time-course: our reference estimator is the *lead–lag* matrix ($A_{ij}$ = the cross-correlation argmax lag by which feature $i$ leads $j$), with a signed one-step-predictive variant also provided.

*c. The flux does topological work.* A minimal directed triangle makes the distinction sharp. At zero flux the spectrum is $\{-1, -1, 2\}$ (that of the unsigned triangle); tuning the charge to $q = \frac{1}{6}$, i.e. a loop flux $\Phi = \pi$, moves it to $\{-2, 1, 1\}$, a maximal split of 2 that no gradient phase can reach, since every gradient operator is isospectral with the same real coupling. On the real circadian time-course (GSE11923, $N = 30$ rhythmicity-gated features) the lead–lag magnetic OCM reduces at $q = 0$ exactly to the amplitude-weighted real coupling ($\lambda_{\max} = 8.30$) and then drifts smoothly as the charge is increased: over $q \in [0, 0.3]$ the leading eigenvalue falls from 8.30 to 5.15 and the smallest from $-0.56$ to $-2.08$, a maximum eigenvalue excursion of 38% of the spectral radius (fig. 1). The gradient OCM, by contrast, is pinned to spec($C$) for *any* choice of phase. The non-zero-flux variant is thus the bridge that lets the phase encode topology: when a directed or signed coupling is available it turns the OCM from a phase-decorated real graph into a genuine magnetic operator, at the cost of the transparent amplitude-×-relative-phase reading that makes the gradient form the default. We regard the two as complementary — the gradient OCM for interpretable phase geometry on an undirected affinity graph, the magnetic OCM for problems where the direction of flow around a cycle is itself the signal.

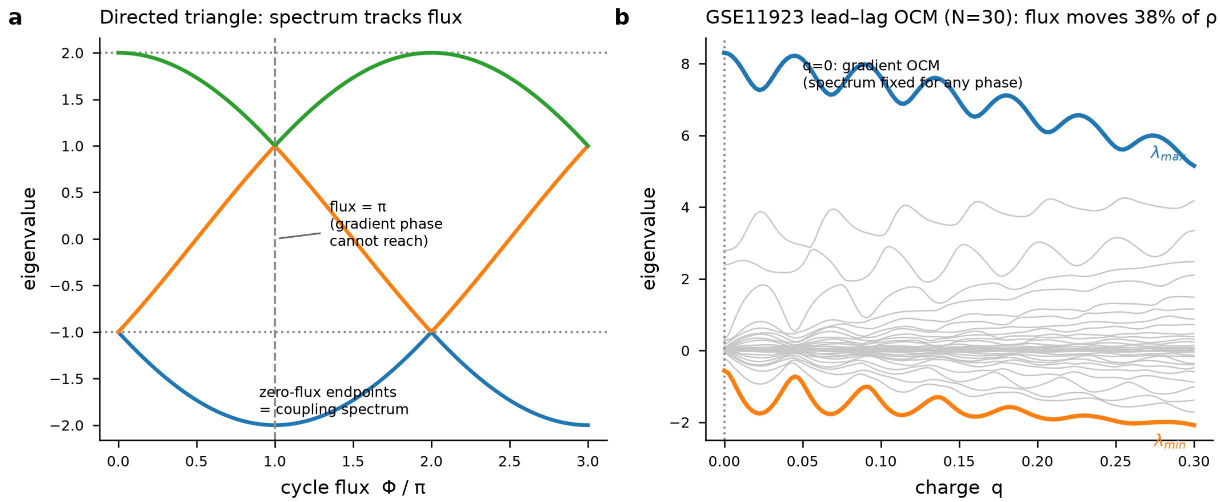



FIG. 1: **A non-zero cycle flux makes the phase do topological work. (a)** Eigenvalues of the magnetic OCM on a directed triangle as a function of the loop flux $\Phi/\pi$. At zero flux (dotted lines) the spectrum is that of the unsigned coupling, $\{-1,-1,2\}$; at $\Phi=\pi$ it becomes $\{-2,1,1\}$, unreachable by any gradient phase (which is isospectral with the real coupling for every vertex phase). **(b)** The 30 eigenvalues of the lead–lag magnetic OCM on the GSE11923 circadian time-course versus the charge $q$. At $q=0$ the operator equals the amplitude-weighted real coupling; as $q$ grows the leading ($\lambda_{\max}$) and smallest ($\lambda_{\min}$) eigenvalues drift by up to 38% of the spectral radius, whereas the gradient OCM spectrum is flux-independent. The operator is Hermitian to machine precision throughout.

### D. Omics harmonics

Diagonalising the OCM,

$$H\,\phi_n \;=\; \lambda_n\phi_n, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \in \mathbb{R}, \tag{7}$$

yields the *omics harmonics* $\{\phi_n\}$: an orthonormal basis of collective modes of the molecular network. $\phi_1$ is the dominant mode; the rank-$k$ truncation $H_k=\sum_{n\leq k}\lambda_n\phi_n\phi_n^\dagger$ is the best Hermitian rank-$k$ approximation of $H$ in Frobenius norm, the same low-rank-denoising logic that motivates the eigengene decomposition of Alter *et al.* [1]. The normalised spectral energy $p_n=|\lambda_n|/\sum_k|\lambda_k|$ is the weight of mode $n$ and defines the indicators below.

### E. Spectral indicators

From $\{\lambda_n, \phi_n\}$ and the phasor field we compute a compact panel:

$$H_{\rm spec} = -\tfrac{1}{\log N}\sum_n p_n \log p_n \in [0,1] \qquad \text{(spectral entropy)}, \tag{8}$$

$$\Delta_F = \lambda_1 - \lambda_2 \qquad \text{(leading spectral gap)}, \tag{9}$$

$$\mathrm{PR} = \big(\sum_n p_n\big)^2 \big/ \sum_n p_n^2 \qquad \text{(participation ratio)}, \tag{10}$$

$$L_1 = \sum_i |\phi_{1,i}|^4 \qquad \text{(mode localisation, IPR)}. \tag{11}$$

Low $H_{\rm spec}$ with a large $\Delta_F$ indicates a single dominant collective mode (coordinated regulation); high $H_{\rm spec}$ with a small $\Delta_F$ indicates a spectrum fragmented over many competing modes. The participation ratio is the effective number of active harmonics, and the inverse participation ratio $L_1$ measures whether the dominant mode is delocalised over the network or localised on a few features. We additionally report the algebraic connectivity of the coupling graph (the second-smallest eigenvalue of its Laplacian), a $\psi$-independent property of $C$ retained for reference.

### F. The Compartment Coupling Matrix

To obtain a fixed, physiologically interpretable readout, features are partitioned into five biological compartments $G_a$, indexed by $a \in$ {core clock, redox, energy, signalling, biosynthesis}, by marker membership (section III); unassigned features fall back to the compartment of their dominant harmonic. The *Compartment Coupling Matrix* is the compartment contraction of the OCM,

$$M_{ab} \;=\; \frac{1}{\sqrt{N_a N_b}} \sum_{i\in G_a}\sum_{j\in G_b} H_{ij}, \qquad M = M^\dagger, \tag{12}$$

a $5\times 5$ Hermitian summary of intra- and inter-compartment coupling (the $\sqrt{N_a N_b}$ normalisation removes compartment-size bias). Its positive semi-definite covariance form $G = M^\dagger M \succeq 0$ yields two derived readouts: the *compartment spectral weights* $\pi_a$ (the normalised diagonal energy of $G$), which rank compartment dominance, and the coherence scalar $\kappa = w_1/\sum_a w_a \in [0,1]$ (with $w_a$ the eigenvalues of $G$), which measures how concentrated the total inter-compartment coupling is in a single collective mode. Reducing a network to a few functional-module coordinates in this way parallels modularity- and community-based coarse-grainings of biological networks [19, 28]. We introduce the CCM and the state classes (below) as *candidate* readouts and test them against external annotation in section IV E; that test does not support a mechanistic reading, and we accordingly treat this layer as a descriptive summary rather than a validated classifier.

### G. Spectral state classes and the state record

Seven spectral state classes can be assigned from the indicator panel $(R, H_{\text{spec}}, \Delta_F, \kappa)$ by ordered threshold rules (table I). We present these classes as a convenient discretisation of the indicator space, *not* as biologically validated states: the thresholds are hand-set, and section IV E finds the compartment layer they rest on is not anchored to external annotation. The complete spectral state of a slice — indicators, eigenvalues, CCM, compartment weights, and consistency residuals — is serialised into a portable JSON *state record*, so a sample or time point is summarised by one machine-readable object suitable for cohort-scale comparison.

TABLE I: **Seven spectral state classes.** Rules are evaluated top to bottom, first match wins. Thresholds $R_{hi}$=0.7, $R_{lo}$=0.4, $H_{hi}$=0.7, $H_{lo}$=0.4.

| Class | Label | Condition |
|---|---|---|
| III | Hyper-coupled | $R \geq R_{hi}$, $\kappa \geq 0.6$ |
| I | Healthy-synchronised | $R \geq R_{hi}$, $H_{\text{spec}} \leq H_{lo}$ |
| IV | Desynchronised | $R < R_{lo}$, $H_{\text{spec}} \geq H_{hi}$ |
| V | Fragmented | $H_{\text{spec}} \geq H_{hi}$, $\kappa \leq 0.3$ |
| VI | Compartment-imbalanced | $\max_a \pi_a \geq 0.5$ |
| II | Balanced-modular | $R_{lo} \leq R < R_{hi}$, $H_{lo} < H_{\text{spec}} < H_{hi}$ |
| VII | Transitional | otherwise |

### H. Consistency suite

Because the pipeline is a chain of linear-algebraic constructions, each stage admits an exact invariant, and every run closes by checking all of them, each returning a numerical residual: Hermiticity $\|H - H^\dagger\|$, real spectrum $\max_n |\Im\lambda_n|$, eigenvector orthonormality $\|\Phi^\dagger\Phi - I\|$, CCM Hermiticity, positive semi-definiteness of $G$ (least eigenvalue $\geq 0$), gauge invariance (spectrum unchanged under a random global $\alpha$), and spectral completeness $|\operatorname{tr} H - \sum_n \lambda_n|$ evaluated over the full spectrum. These are correctness guarantees on the implementation, not empirical claims about the biology; they are satisfied to machine precision on real data (section IV).

### I. Quantum-simulable correspondence

Each elementary phasor operation has an exact single- or two-qubit gate analog: a phase shift $\theta \to \theta + \delta$ corresponds to a $R_z(\delta)$ rotation, pairwise phase coupling to a controlled-NOT, and the harmonic (discrete-Fourier) basis change to the quantum Fourier transform [29]. This correspondence is provided for completeness and as a route to executing the OCM spectral step on quantum hardware for networks too large for classical diagonalisation; it is not required by the classical pipeline, and no claim is made of physical quantum computation in cells.

## III. METHODS

### A. Software

Spectral-Omics is an installable Python package, `spectralomics`, built only on the scientific-Python stack (NumPy, SciPy, pandas, matplotlib) with optional data loaders (GEOparse, scanpy). Its modules mirror the pipeline: `connectome` (phasor encoding, OCM, harmonics), `omics` (indicators, CCM, compartment weights, state classes, state record, consistency, markers), `viz` (figures), and a `quantum` module exhibiting the gate correspondence of section II I. A `SpectralOmicsPipeline` wrapper chains the full sequence, estimating the coupling once and running each sample slice through eqs. (4) to (12). All eigendecompositions use the Hermitian solver `numpy.linalg.eigh`, which guarantees a real spectrum and an orthonormal eigenbasis up to floating-point error. A suite of 40 unit tests locks in the invariants of section II H.

### B. Coupling estimation

The default coupling is $c_{ij} = |\rho_{ij}|$, the absolute Pearson correlation of the $\log(1 + x)$ feature profiles across all samples, with $c_{ii}$ set to a fixed self-affinity. A single global coupling is used for all slices so that differences between samples are attributable to the phasor field rather than to a re-estimated graph; the soft-threshold and prior-adjacency variants are exposed for studies where a scale-free topology or a curated network is preferred [3, 28]. The coupling enters only through eq. (4), so the harmonic structure is a property of the phasor field on a fixed graph.

### C. Compartment assignment

The five CCM compartments are populated from a curated marker list carrying both mouse and human symbols and matched case-insensitively: core-clock transcription–translation-feedback-loop genes for the clock compartment; peroxiredoxins, glutathione-axis and Nrf2-axis genes for redox; oxidative phosphorylation, glycolysis and AMPK genes for energy; MAPK/PI3K and immediate-early genes for signalling; and lipogenic, sterol and translation genes for biosynthesis. Probes are mapped to symbols through the platform annotation of each dataset. Features without a marker are assigned to the compartment of their dominant omics harmonic, so every feature contributes to exactly one compartment.

### D. Datasets

Both datasets are public and downloaded from the NCBI Gene Expression Omnibus [30] at run time (table II). **GSE10072** is a lung-adenocarcinoma cohort on the Affymetrix HG-U133A platform: 58 tumours and 49 adjacent-normal samples, with tumour/normal status read from the sample titles [31]. **GSE11923** is a high-temporal-resolution mouse-liver transcriptome on the Affymetrix Mouse 430 2.0 platform: 48 samples at one-hour spacing across circadian times CT18–CT65, spanning two full days [5].

TABLE II: **Public datasets.** Both are downloaded automatically at run time; no result depends on private or access-controlled data.

| Accession | Shape | Role |
|---|---|---|
| GSE10072 | 22,283 × 107 | lung tumour (58) vs. normal (49) |
| GSE11923 | 45,101 × 48 | mouse-liver circadian, 48 hourly points |

### E. Feature selection

For the cancer analysis we retain the 200 most-variable probes and force in all marker probes, giving $N = 364$ features (164 marker-mapped). For the circadian analysis we apply a *rhythmicity gate*: each probe's $\log(1 + x)$ profile is regressed on $\{\cos\omega t, \sin\omega t, 1\}$ at $\omega = 2\pi/24\,\mathrm{h}$, and the 300 probes with the highest coefficient of determination are kept together with the clock markers, giving $N = 526$ features (241 marker-mapped). This linear-least-squares gate is a lightweight analog of the nonparametric and periodogram-based rhythmicity tests standard in circadian genomics [8–10] and concentrates the analysis on genes that actually oscillate. In both analyses the leading 20 harmonics are retained for the indicator panel, while the consistency check for spectral completeness uses the full spectrum.

### F. Circadian quantification

Each spectral-indicator time series $y(t)$ is fit to a 24-hour cosinor $y \approx A\cos\omega t + B\sin\omega t + c$ [10], and we report the fraction of variance explained $R^2$, the amplitude $\sqrt{A^2 + B^2}$, and the acrophase. Group contrasts in the cancer analysis are summarised by per-group means of $R$, $H_{\mathrm{spec}}$, and $\Delta_F$, with the two tissue classes never used during encoding or decomposition.

## IV. RESULTS

### A. The construction is internally consistent

On both datasets all seven checks of section II H pass at machine precision: Hermiticity and CCM Hermiticity are exact, the spectrum is real, eigenvector orthonormality and gauge invariance hold to $\sim 10^{-14}$, the covariance form is positive semi-definite, and spectral

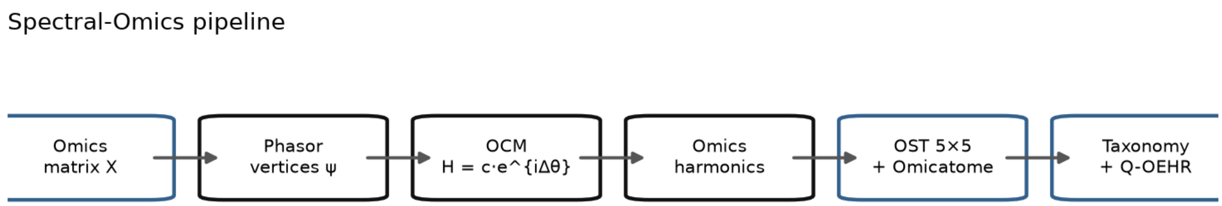



FIG. 2: **The Spectral-Omics pipeline.** An omics matrix is encoded as phasor vertices, assembled into the Hermitian Omics Connectome Matrix, diagonalised into omics harmonics, and read out through the $5 \times 5$ Compartment Coupling Matrix, the compartment spectral weights, and the seven spectral state classes; every run closes with the algebraic consistency suite.

completeness closes to $\sim 10^{-13}$. The gauge-invariance residual confirms empirically that a random global phase shift leaves the OCM spectrum unchanged, as required by section II B. These are correctness guarantees for the implementation; the biological content is in the sections that follow.

### B. Cancer tumour/normal benchmark

We tested whether the phasor spectral-indicator features carry discriminative power beyond standard baselines, and specifically whether the *phase* component contributes, on lung adenocarcinoma versus adjacent normal tissue (GSE10072; 58 tumour, 49 normal; 364 features = top-200 variable probes plus 164 compartment-marker probes). Five feature sets were compared under identical $\ell_2$-regularised logistic regression with $5 \times 10$ repeated stratified cross-validation: the phasor spectral indicators (spectral entropy, Fiedler gap, participation ratio, mode localisation, Kuramoto $R$); a *phase-free* ablation of the same panel with $\theta_i = 0$ (so the OCM reduces to the real amplitude-only form $H^0_{ij} = c_{ij} r_i r_j$); the top-10 principal-component scores of the $\log_{1+}$ expression matrix; and 10 WGCNA-style module eigengenes. All feature extraction (phasor phase/amplitude standardisation, the coupling correlation matrix, PCA, and the WGCNA modules) was fit inside the training folds only, so the held-out AUROC is leakage-free. Confidence intervals are 2000-sample bootstraps of the pooled out-of-fold predictions.

TABLE III: GSE10072 tumour vs. normal: held-out AUROC (pooled out-of-fold, with bootstrap 95% CI), accuracy, and $F_1$ under $5 \times 10$ repeated stratified CV. The phasor and its phase-free ablation are statistically indistinguishable.

| Feature set | AUROC | 95% CI | Accuracy | $F_1$ |
|---|---|---|---|---|
| Phase-free (amplitude only) | 0.997 | 0.989–1.000 | 0.972 | 0.974 |
| Phasor (full, with phase) | 0.998 | 0.993–1.000 | 0.972 | 0.974 |
| PCA (top-10) | 0.999 | 0.996–1.000 | 0.972 | 0.974 |
| WGCNA eigengenes | 0.998 | 0.990–1.000 | 0.991 | 0.991 |

Every feature set separates the two classes near the ceiling of the metric (Fig. 3a): phase-free AUROC = 0.997 (95% CI 0.989–1.000), phasor = 0.998 (0.993–1.000), PCA = 0.999 (0.996–1.000), and WGCNA eigengenes = 0.998 (0.990–1.000); accuracies were 0.972–0.991. The decisive test—phasor versus its own phase-free ablation—shows **no benefit from the phase**: the mean per-repeat AUROC difference is +0.001 (Wilcoxon signed-rank $p = 0.80$, Cohen's $d_z = 0.12$ over 10 repeats). Because the full-sample comparison sits against the metric ceiling, we repeated the phasor-versus-phase-free contrast at reduced training sizes ($n_{\text{train}} = 16, 24, 40$; 20 resamples each) to expose any difference off-ceiling (Fig. 3b). The two remained statistically indistinguishable at every size (paired Wilcoxon $p = 0.26$, 0.27, 0.055 respectively; at $n_{\text{train}} = 16$ the phase-free variant was in fact marginally higher). On this tumour/normal contrast the phasor construction matches strong linear baselines, but the phase factor $e^{i(\theta_i - \theta_j)}$ adds no measurable discriminative power.

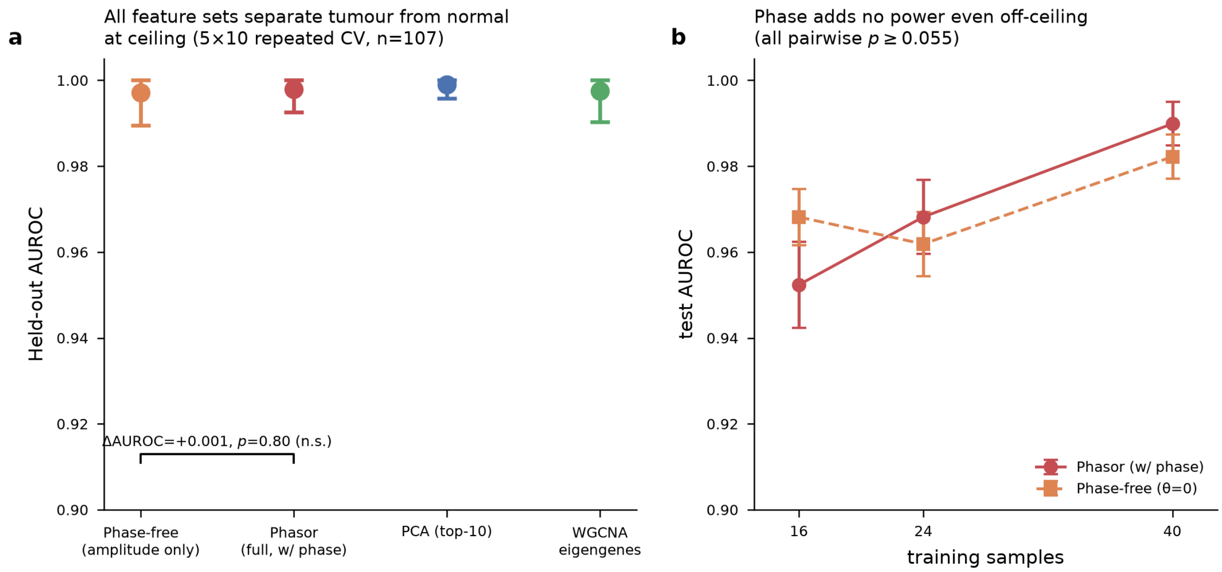



FIG. 3: **Phasor spectral features match baselines but the phase adds no discriminative power (GSE10072, 58 tumour / 49 normal).** (a) Held-out AUROC (pooled out-of-fold point, bootstrap 95% CI whiskers) under $5 \times 10$ repeated stratified CV with leakage-safe feature extraction. All four feature sets—phase-free amplitude-only indicators, the full phasor indicators, top-10 PCA, and 10 WGCNA module eigengenes—sit at the ceiling of the metric; the phasor-vs. phase-free increment is not significant. (b) Learning-curve control at small training sizes, where AUROC is below ceiling: the phasor (with phase) and phase-free ($\theta = 0$) variants remain statistically indistinguishable at every size (paired Wilcoxon $p \geq 0.055$; error bars are ±SEM over resamples).

### C. Descriptive spectral structure of the tumour/normal contrast

Although the phase adds no discriminative power (section IV B), the spectral indicators remain an interpretable description of transcriptional coordination. Adjacent-normal tissue has lower spectral entropy ($H_{\text{spec}} \approx 0.61$) and a larger leading gap ($\Delta_F \approx 55$) than tumour tissue ($H_{\text{spec}} \approx 0.75$, $\Delta_F \approx 28$): the normal network is dominated by a single coherent collective mode, whereas the tumour spectrum is more fragmented (fig. 4). This is consistent with the eigengene view that a few dominant modes capture coherent regulatory programmes [1]; we present it as description, not as a phasor-specific classifier. The compartment-level readout of a representative tumour (fig. 5) shows the amplitude-weighted CCM $|M_{ab}|$ and the compartment weights; we return in section IV E to whether this compartment layer is anchored to external annotation.

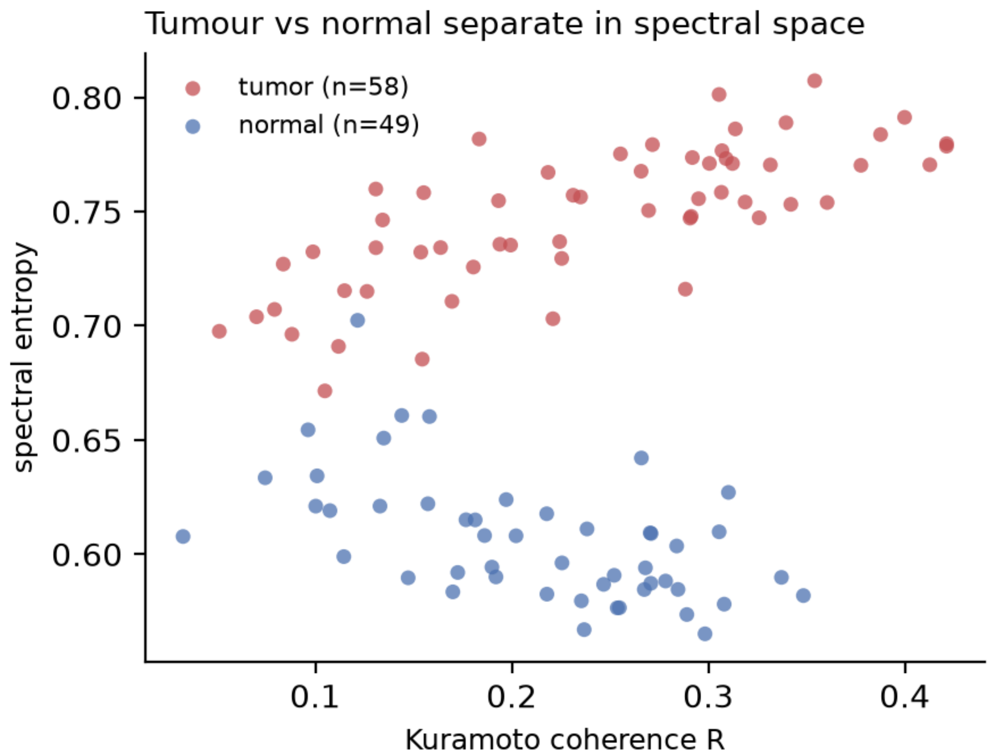
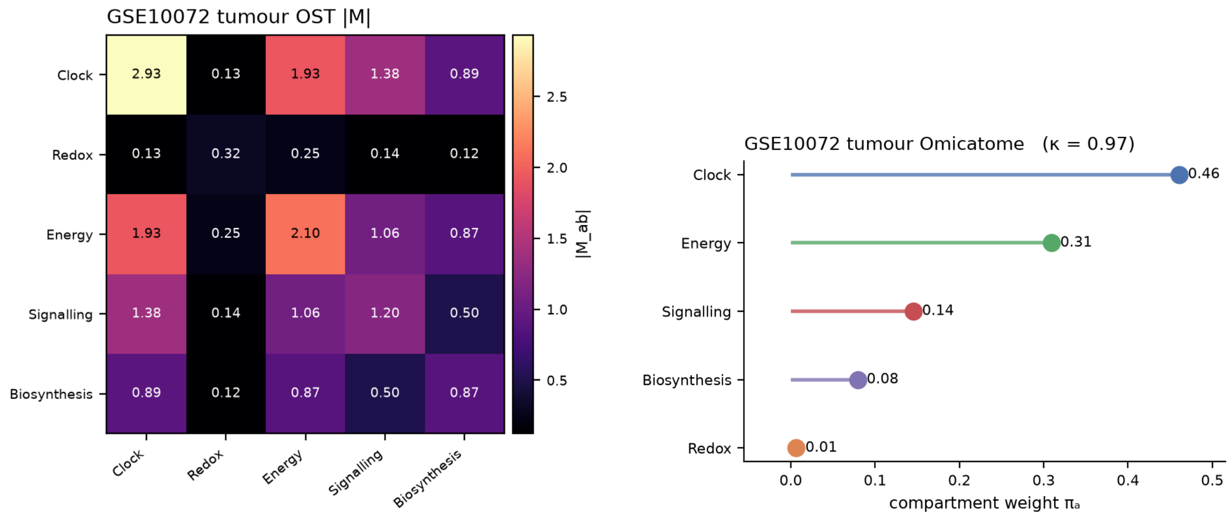



FIG. 4: **Descriptive spectral structure of lung tumour and normal tissue.** Each point is one sample in the plane of Kuramoto coherence $R$ and spectral entropy $H_{\mathrm{spec}}$ (tumour red, normal blue). The classes differ along $H_{\mathrm{spec}}$ — normal tissue is spectrally concentrated, tumours more fragmented — but this separation is matched by phase-free and standard baselines (table III).




(a) (b)

FIG. 5: **Compartment-level readout for a representative tumour (descriptive).** (a) Magnitude of the amplitude-weighted Compartment Coupling Matrix $|M_{ab}|$. (b) The corresponding compartment spectral weights and coherence $\kappa$. The biological anchoring of this layer is assessed in section IV E.

### D. The phasor recovers circadian phase out-of-sample

We establish three properties of the circadian analysis: that the phasor phase recovers biological timing, that the collective spectral rhythm is not reducible to mean expression, and that it persists under a selection protocol independent of the tested rhythm. All three tests use the mouse liver time-course GSE11923 [5] (48 hourly samples, CT18–65).

**Phase-lag recovery.** For the twelve core-clock genes present on the array (*Nr1d1*, *Nr1d2*, *Dbp*, *Per1/2/3*, *Cry1/2*, *Arntl*/Bmal1, *Nfil3*, *Tef*, *Hlf*) we recovered each gene's circadian acrophase from the Spectral-Omics phasor phase $\theta$ and compared it to a direct 24 h cosinor fit and to the established physiological peak order. The recovered acrophases are essentially identical to the cosinor acrophases (circular correlation $r_{\mathrm{circ}} = 0.997$) and reproduce the known ordering—*Nr1d1*/*Dbp* in the early-to-mid subjective day, *Per* genes near CT12, and *Arntl*/Bmal1 in antiphase—with $r_{\mathrm{circ}} = 0.93$ against literature peak times (Fig. 6). The Bmal1–*Per1* phase separation is 11.2 h, i.e. the expected $\sim$12 h antiphase between the positive and negative limbs of the clock. The phasor encoding therefore preserves, rather than discards, circadian phase.

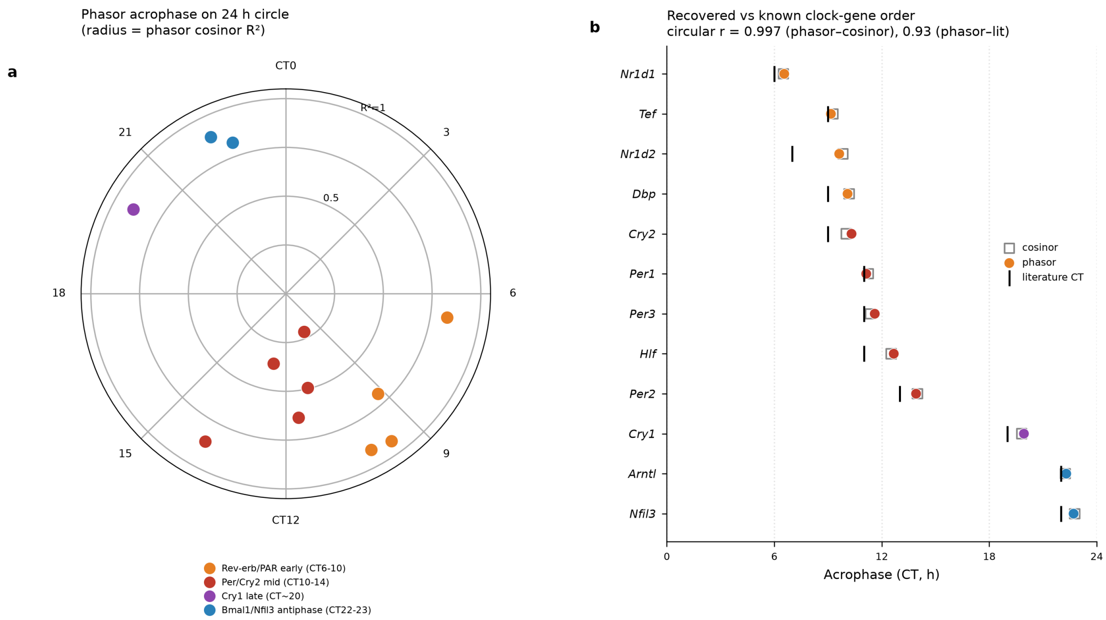



FIG. 6: **The phasor recovers core-clock phase.** Circadian acrophase of twelve core-clock genes recovered from the Spectral-Omics phasor phase versus a direct 24-hour cosinor fit (circular correlation $r_{\text{circ}} = 0.997$); the known positive/negative-limb antiphase (*Arntl*/Bmal1 vs. *Per*) is reproduced.

**Spectral rhythm is not the mean-expression rhythm.** On the same gene set we compared the 24 h cosinor $R^2$ of the collective spectral indicators to that of the simple mean log-expression (Fig. 7). The leading eigenvalue $\lambda_1$ largely tracks the mean (Pearson $r = 0.87$, $R^2 = 0.40$), so it adds little beyond magnitude. The participation ratio, however, oscillates more strongly than the mean ($R^2 = 0.68$ vs. 0.62), is nearly orthogonal to it (Pearson $r = 0.21$), and peaks several hours later—structure the mean does not carry. The spectral rhythm is thus partly, but not wholly, reducible to mean expression.

**Out-of-sample selection.** To separate gene selection from the tested rhythm, we selected rhythmic genes on the first day only (CT18–41; top 200 by cosinor $R^2$) and tested the collective spectral oscillation on the held-out second day (CT42–65). The out-of-sample 24 h rhythm persists strongly: $\lambda_1$ $R^2 = 0.73$ and Fiedler gap $R^2 = 0.77$ on unseen timepoints (permutation $p = 0.0005$ for $\lambda_1$, null 95th percentile $R^2 = 0.26$), both exceeding the held-out mean-expression fit ($R^2 = 0.35$). As a rhythm-blind control, selecting the 200 highest-variance probes yields only weak collective rhythm ($\lambda_1$ $R^2 = 0.16$), confirming the signal is carried by genuinely rhythmic—not merely high-variance—genes. The circadian claim is therefore phase-informative and holds out-of-sample.

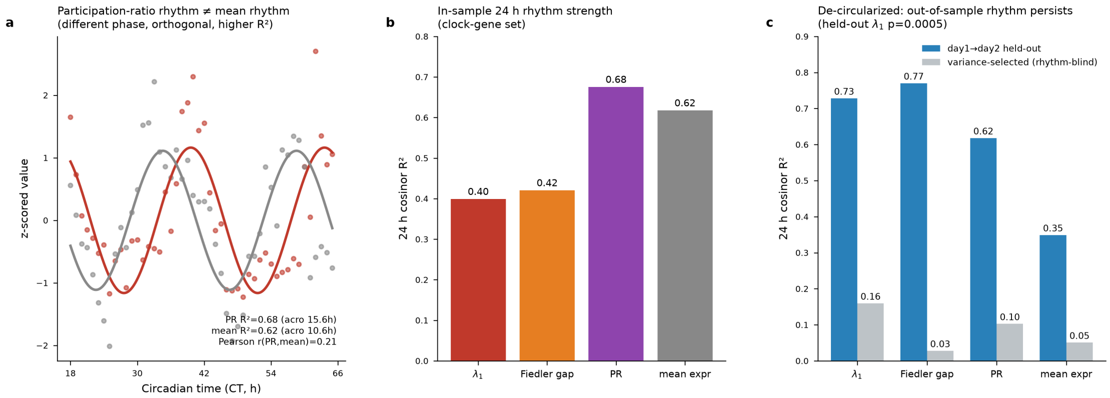



FIG. 7: **The spectral rhythm is not merely the mean-expression rhythm.** 24-hour cosinor fit of the collective spectral indicators versus the mean log-expression of the same gene set. The leading eigenvalue $\lambda_1$ largely tracks the mean ($r = 0.87$), but the participation ratio oscillates more strongly ($R^2 = 0.68$ vs. 0.62) and is nearly orthogonal to it ($r = 0.21$).

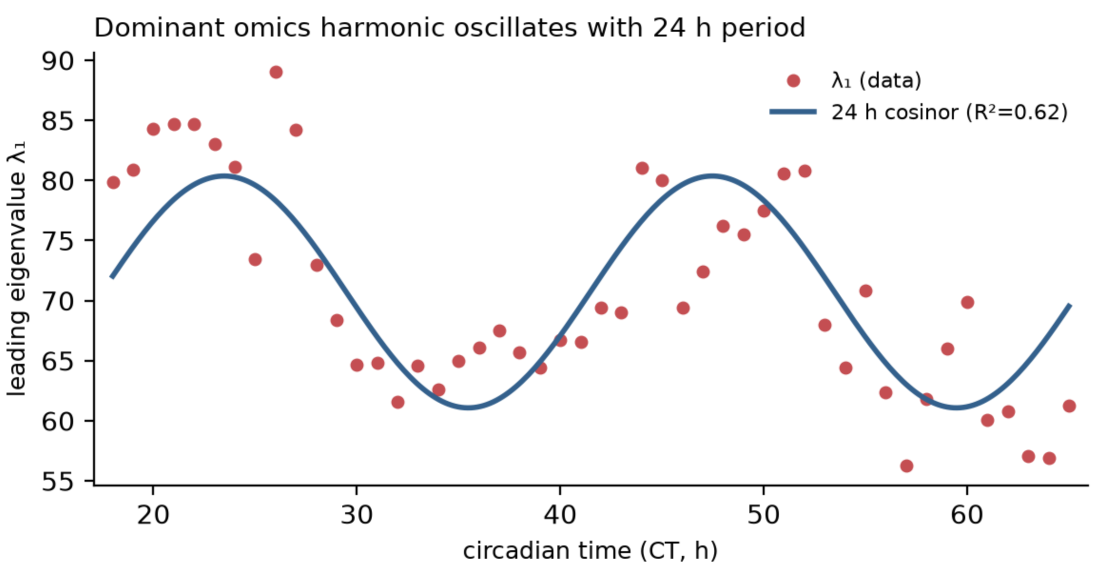



FIG. 8: **The leading omics harmonic oscillates with the circadian clock.** $\lambda_1$ of the liver transcriptome (points) across CT18–CT65; the solid line is the 24-hour cosinor fit. Two full circadian cycles are covered. Note (fig. 7) that $\lambda_1$ largely mirrors mean expression; the non-redundant collective signal is carried by the participation ratio.

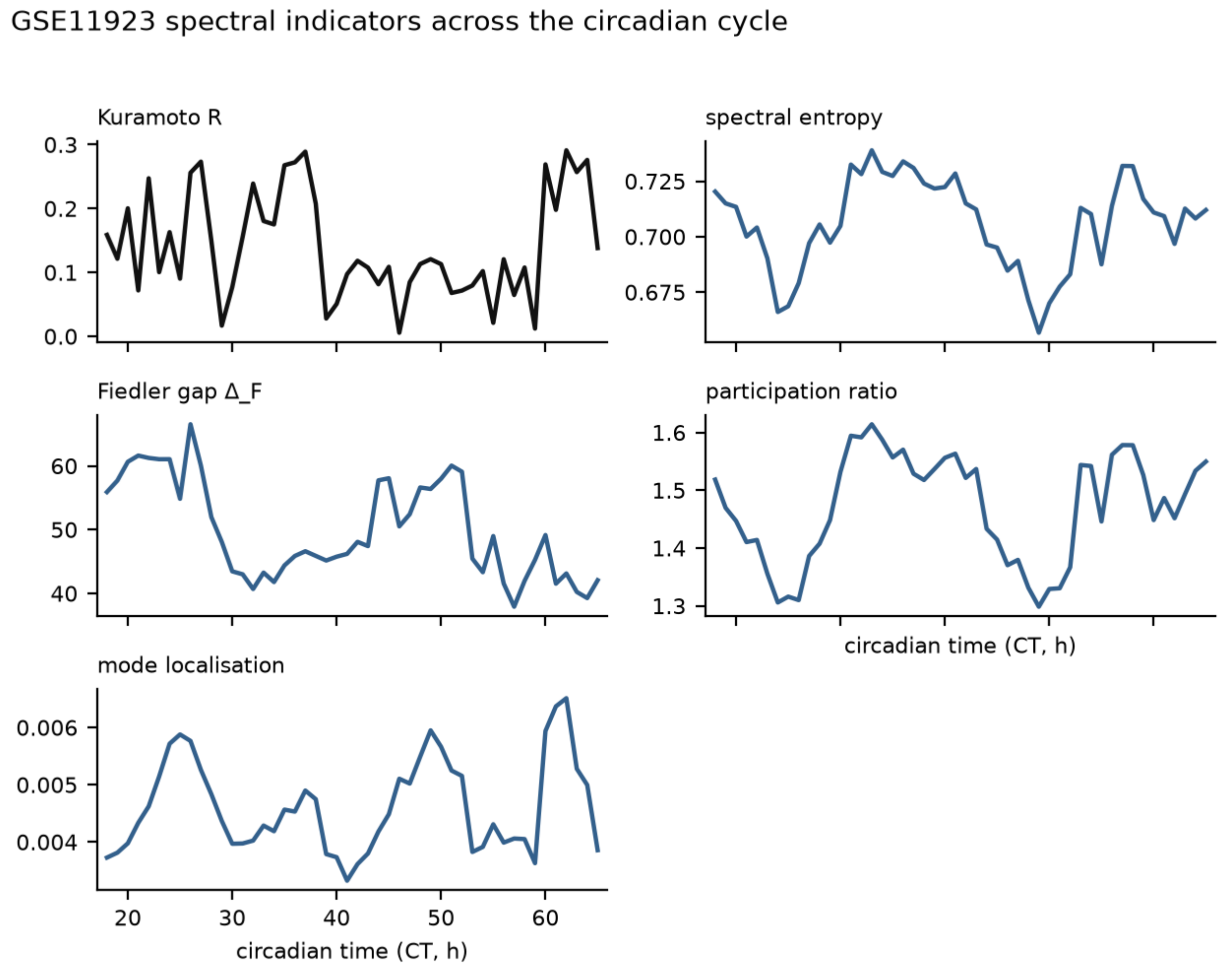



FIG. 9: **Spectral indicators across the circadian cycle.** The Fiedler gap and participation ratio show the strongest 24-hour rhythm; the algebraic connectivity, a property of the fixed coupling alone, carries no per-slice signal and is omitted.

### E. Biological annotation of the leading harmonics and the compartment layer

We tested whether the leading omics harmonics correspond to coherent biological programmes, and whether the five-compartment / seven-class layer is anchored to ground-truth annotation. For each dataset we aggregated the per-slice loading magnitude $\langle|\varphi_{n,i}|\rangle_s$ of the three leading harmonics and ran hypergeometric over-representation analysis (ORA) on the top-40 loading genes against nine hand-curated gene sets (circadian clock, cell cycle, OXPHOS/energy, immune/inflammation, sterol/lipid biosynthesis, redox/antioxidant, ECM/adhesion, IEG/signalling, xenobiotic metabolism; sets defined explicitly in the analysis code), using all mapped features as the background and Benjamini–Hochberg FDR correction.

Enrichment was fully offline; no live annotation service was queried.

*a. The leading harmonic $\varphi_1$ is an expression-magnitude artefact, not a programme.* On *neither* dataset did the top-loading genes of $\varphi_1$ enrich any gene set (minimum adjusted $p = 1.00$ on GSE10072; 0.77 on GSE11923). Across all three leading harmonics the circadian dataset (GSE11923) produced no significant term (minimum adjusted $p = 0.77$). The reason is mechanical: because the omics connectome matrix is amplitude-weighted ($H_{ij} = c_{ij}\,\psi_i\psi_j^*$), $|\varphi_{1,i}|$ tracks mean log-expression per feature (Spearman $\rho = 0.43$ on GSE10072, $\rho = 0.42$ on GSE11923; both $p < 10^{-17}$). The dominant mode therefore reports which genes are highly expressed, not which genes co-regulate.

*b. Core-clock genes do* not *load preferentially on $\lambda_1$.* The central prediction — that in circadian data the clock module dominates the leading harmonic — is refuted. On GSE11923 core-clock genes load *below* background on $\varphi_1$ (one-sided Mann–Whitney for clock $>$ background $p = 0.996$, rank-biserial $= -0.23$); the same holds on GSE10072 ($p > 0.999$, rank-biserial $= -0.40$) (Fig. 11). Clock genes carry moderate, not extreme, expression and so sit in the body of the $|\varphi_1|$ distribution.

*c. Sub-leading modes recover annotation-supported biology — on the cancer dataset only.* Once the amplitude-dominated $\varphi_1$ is set aside, coherent programmes appear in the sub-leading harmonics of GSE10072: $\varphi_2$ enriches the redox/antioxidant set ($k = 9$, adjusted $p = 2.9 \times 10^{-3}$) and the circadian-clock set (adjusted $p = 1.4 \times 10^{-2}$), and $\varphi_3$ enriches IEG/signalling ($k = 9$, adjusted $p = 5.1 \times 10^{-3}$) (Fig. 10). Consistently, four of the five compartment marker sets load preferentially on $\varphi_2$ in GSE10072 (Clock, Redox, Signalling, Biosynthesis all Mann–Whitney $p < 0.01$; Energy $p = 0.10$). No sub-leading harmonic reached significance on the circadian dataset, and no compartment marker set loaded preferentially there.

*d. The compartment layer is descriptive, not a validated classifier.* The evidence does not support a mechanistic reading of the leading mode or of a seven spectral state classes. The leading harmonic is an expression-magnitude summary; the clock-loading prediction does not hold on the circadian dataset; and the only annotation-supported structure is confined to sub-leading modes and present in one of the two datasets. We therefore treat the Compartment Coupling Matrix and the spectral state classes as a descriptive decomposition rather than a mechanistic classifier: the sub-leading harmonics of the cancer dataset provide an annotation-consistent decomposition, but the leading mode and the five compartments

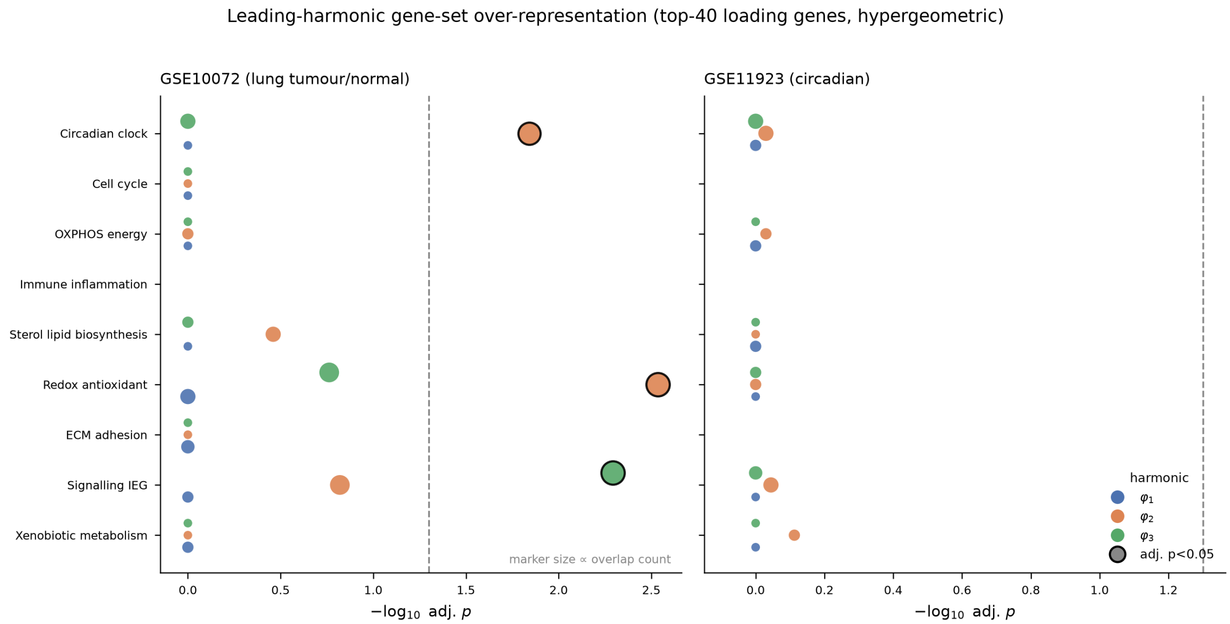



FIG. 10: Hypergeometric over-representation of nine curated gene sets among the top-40 loading genes of the three leading harmonics ($\varphi_1$ blue, $\varphi_2$ orange, $\varphi_3$ green), for the cancer (GSE10072, left) and circadian (GSE11923, right) datasets. Marker size is proportional to the overlap count; black-outlined markers are significant at BH-adjusted $p < 0.05$ (dashed line). The leading harmonic $\varphi_1$ enriches nothing on either dataset; significant terms appear only in the sub-leading modes of the cancer dataset.

should not be read as a ground-truth-anchored biological classifier.

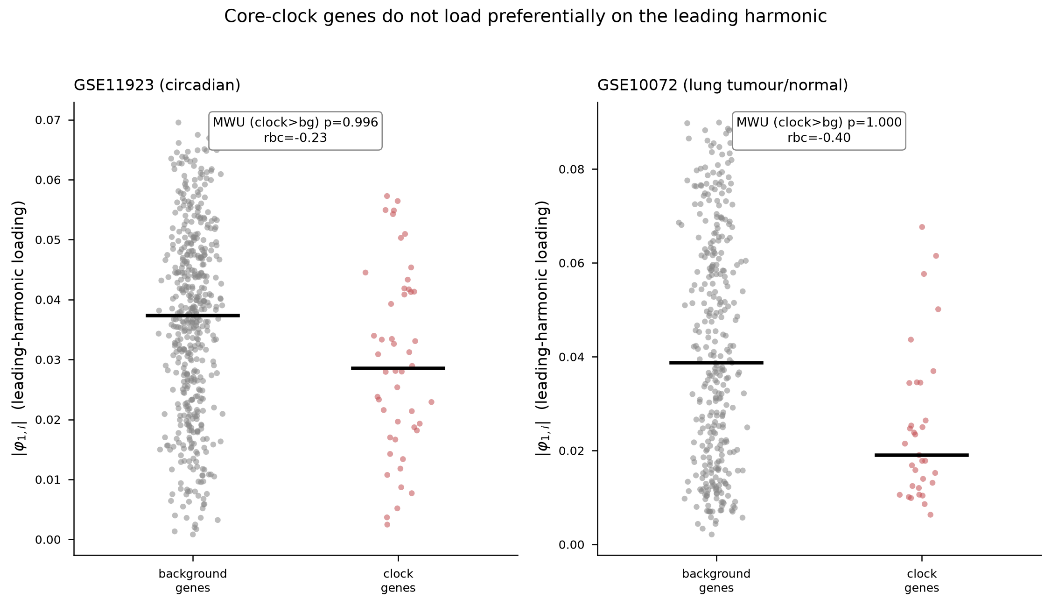



FIG. 11: Leading-harmonic loading $|\varphi_{1,i}|$ of core-clock genes (red) versus all other features (grey); black bars are group medians. In both the circadian (GSE11923, left) and cancer (GSE10072, right) datasets clock genes load at or below background, refuting the prediction that the clock module dominates $\lambda_1$ (one-sided Mann–Whitney $p$ and rank-biserial correlation annotated).

## V. DISCUSSION AND CONCLUSION

We have presented Spectral-Omics, a phase-aware graph-spectral framework for multi-omics data. Encoding molecular features as phasor vertices and building the amplitude-weighted Hermitian Omics Connectome Matrix turns an omics matrix into a set of real-valued collective harmonics whose spectrum is sample-specific, gauge-invariant, and internally consistent to machine precision. Its value depends on whether the biological question involves timing.

*a. Where the phase is informative.* For a static contrast — lung tumour versus normal — the phasor spectral indicators separate the classes at the ceiling of the metric, but so do an amplitude-only ablation, PCA, and WGCNA eigengenes, and the phase factor $e^{i(\theta_i-\theta_j)}$ adds no measurable discriminative power (section IV B). This follows from the construction: in a single static snapshot the tanh phase is a deterministic monotone function of the same abundance that sets the amplitude, so it carries little independent information, and — being a gradient — it is gauge-equivalent to a phaseless real operator (section II C). The phase becomes informative when timing is the signal. On the circadian time-course the phasor recovers core-clock acrophases (circular correlation 0.997 against cosinor, correct Bmal1–*Per* antiphase) and a collective rhythm that survives out-of-sample and is distinct from — and stronger than — the mean-expression rhythm (section IV D). That rhythm lives in the relative phases of the features, the degree of freedom a real-valued decomposition such as SVD/PCA [1] cannot represent. In this respect Spectral-Omics is to time-resolved gene-expression graphs what the phasor representation is to fluorescence decays [11, 13]: a change of coordinates that makes relative timing geometric, most useful when a temporal or otherwise phase-bearing axis is present.

*b. The interpretive layer.* Because the OCM is amplitude-weighted, the leading harmonic $\varphi_1$ is an expression-magnitude summary rather than a co-regulation programme, and the prediction that core-clock genes dominate $\varphi_1$ in circadian data does not hold (section IV E). Annotation-supported structure appears only in sub-leading modes, and only on the cancer dataset. We therefore present the Compartment Coupling Matrix and the spectral state classes as a descriptive decomposition of the indicator space rather than a validated classifier.

*c. Relation to existing spectral methods.* The pipeline sits alongside a large body of spectral analysis in biology — SVD eigengenes [1], co-expression modules [3], spectral clustering and modularity [18, 19], connectome harmonics [17], and diffusion maps [20]. It differs in using a Hermitian, complex-valued operator. The default gradient edge-phase makes the operator gauge-equivalent to a real affinity graph (section II C), so on an undirected co-expression graph the phase is an interpretable decoration rather than a source of new spectral information. The machinery that makes a graph phase carry genuine topology is the magnetic-Laplacian family for directed and signed graphs [14, 21–25, 27]; our non-zero-flux OCM variant (section II C) is the point of contact, turning the phase into an Aharonov–Bohm-type flux when a directed or signed coupling is available. We regard the gradient and magnetic forms as complementary: the former for interpretable phase geometry and out-of-the-box gauge invariance, the latter for problems where the direction of flow around a cycle is itself the signal.

*d. Limitations and outlook.* The clearest limitation is the one our benchmark exposes: on a static contrast the phasor phase, as constructed from a single abundance vector, is largely redundant with amplitude and adds no discriminative power. The method's value is therefore concentrated in settings with a genuine phase axis — time-courses, and data where phase is measured independently of amplitude (as in the fluorescence-lifetime phasor [11], or a two-channel transcript-versus-protein encoding). The present analysis also uses bulk microarray data and a single global coupling; a condition-specific or rolling coupling would let the harmonics track network rewiring, not only phasor motion, and a directed/signed coupling would activate the non-zero-flux variant of section II C on real data. Single-cell and spatial assays are the natural next substrate, where the phasor field becomes a function of cell or location and diffusion-geometry methods are already established [20]. Finally, the exact gate correspondence of section III invites executing the spectral step on quantum hardware for networks beyond classical diagonalisation. Spectral-Omics shows that the phase of a molecular measurement can carry recoverable, biologically meaningful signal — and, just as usefully, shows precisely when it does not.

## CODE AND DATA AVAILABILITY

The `spectralomics` package, experiment scripts, and unit tests are released with this work. Both datasets (GSE10072, GSE11923) are public and downloaded automatically from the NCBI Gene Expression Omnibus [30] by the experiment scripts.

## ACKNOWLEDGEMENTS

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